

NANYANG, China

WMO: 571780

Lat: 33.1008N

Lon: 112.4869E

Elev: 596

StdP: 14.38

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	26.1	0.8	5.9	36.5	6.8	8.0	36.4	16.7	39.0	15.3	38.3	3.3	40	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	95.0	76.4	92.9	76.3	90.8	76.0	82.2	89.9	80.9	88.4	79.7	86.9	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	159.8	87.4	78.9	153.4	85.8	77.7	147.1	84.6	46.8	90.0	45.3	88.7	44.0	86.8	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	12.8	11.2	18.8	98.9	3.5	2.1	16.2	100.4	14.2	101.7	12.2	102.8	9.6	104.4
			16.6	84.0	3.3	1.3	14.2	85.0	12.3	85.8	10.5	86.5	8.1	87.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	36.0	41.6	51.3	62.1	71.6	79.0	81.9	80.3	72.4	62.9	50.3	39.8		
	DBStd	16.97	5.12	6.61	7.28	6.81	5.76	4.99	4.33	5.08	5.56	6.05	6.88	4.99		
	HDD50	1149	434	244	71	4	0	0	0	0	0	1	77	318		
	HDD65	3464	898	656	425	129	12	0	0	1	8	112	441	782		
	CDD50	5120	0	8	113	369	670	870	990	939	672	402	86	1		
	CDD65	1961	0	0	2	44	216	421	525	475	230	48	0	0		
	CDH74	18997	0	0	20	330	1795	4294	5808	4876	1576	292	6	0		
	CDH80	7535	0	0	1	61	628	1862	2479	1962	494	48	0	0		
	WSAvg	4.9	4.8	5.3	5.8	5.5	5.2	4.9	4.7	4.7	4.5	4.1	4.3	4.5		
	PrecAvg	30.0	0.4	0.7	1.3	1.9	3.0	4.1	6.9	4.8	3.3	1.8	1.4	0.4		
(15) Precipitation	PrecMax	45.4	1.5	2.2	2.9	3.6	9.6	9.5	14.6	9.5	8.5	5.1	4.0	1.7		
	PrecMin	21.5	0.0	0.0	0.0	0.2	0.5	0.9	1.1	1.5	0.3	0.1	0.0	0.0		
	PrecStd	5.9	0.4	0.5	0.9	1.0	1.9	2.0	3.3	2.5	2.2	1.5	1.1	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	64.9	76.8	85.5	94.5	97.8	97.0	97.3	93.8	85.4	74.3	59.5		
		MCWB	43.8	50.5	60.2	67.0	67.0	71.2	80.0	79.5	75.2	66.3	57.9	46.2		
	2%	DB	51.7	59.8	71.5	81.3	89.9	94.7	94.5	93.5	88.7	80.7	69.4	55.3		
		MCWB	41.4	48.0	57.3	65.2	67.2	72.5	79.7	78.9	72.9	63.6	55.7	44.3		
	5%	DB	48.3	56.3	67.4	77.9	86.5	92.1	92.4	91.1	85.4	77.3	65.2	51.9		
		MCWB	39.9	45.6	55.1	63.8	67.4	72.5	79.2	77.9	70.9	61.6	53.6	42.6		
	10%	DB	45.2	52.8	63.7	74.1	83.3	89.3	90.3	88.9	82.3	73.7	61.3	48.7		
		MCWB	38.0	44.1	52.9	61.6	66.6	72.0	78.8	77.2	69.4	61.0	52.2	41.5		
	0.4%	WB	45.6	53.6	63.6	72.0	75.2	80.1	84.0	83.5	80.0	69.3	60.7	48.9		
		MCDB	53.2	61.0	72.8	80.5	83.7	89.1	92.1	91.4	89.4	78.9	67.6	54.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.0	49.9	59.8	68.3	72.1	78.4	82.6	82.0	77.2	66.6	57.8	46.6		
		MCDB	49.2	56.8	68.5	76.7	81.4	87.0	90.7	89.1	84.5	74.8	65.8	52.3		
	5%	WB	40.7	47.4	56.9	65.6	70.3	77.0	81.4	80.7	74.4	64.9	55.6	44.5		
		MCDB	46.7	54.1	65.1	74.8	80.6	85.2	89.3	87.5	80.7	72.5	62.6	50.2		
	10%	WB	38.7	45.2	54.0	63.1	68.9	75.5	80.3	79.4	72.0	63.3	53.4	42.3		
		MCDB	44.1	51.7	62.1	71.9	79.1	83.6	87.7	86.0	78.4	70.4	59.8	47.5		
	5% DB	MDBR	13.8	15.3	15.9	16.7	17.3	15.9	12.6	12.6	14.1	15.3	14.7	14.2		
		MCDBR	20.1	21.8	22.0	22.4	23.4	20.9	16.0	15.9	19.4	21.9	21.6	21.2		
		MCWBR	12.5	12.5	11.4	10.5	7.9	6.5	5.7	5.4	6.9	8.3	10.6	12.5		
		MCWBR	18.3	18.9	19.6	19.8	18.1	15.2	13.9	13.1	15.9	16.8	18.1	18.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	12.2	11.5	11.4	10.9	8.2	6.2	6.1	5.7	7.4	7.7	9.9	11.7		
	taub		0.632	0.702	0.702	0.697	0.691	0.769	0.747	0.779	0.736	0.720	0.675	0.615		
		taud	1.547	1.458	1.480	1.525	1.578	1.482	1.555	1.484	1.526	1.500	1.539	1.595		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		173	177	194	206	209	193	196	186	185	171	161	167		
		Edh at Noon	72	87	91	90	86	95	88	93	86	82	72	65		
	RadAvg		704	829	1198	1433	1519	1529	1433	1341	1105	941	757	679		
(45) All-Sky Solar Radiation	RadStd		94	172	128	131	131	143	186	192	144	139	119	112		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	N/A	-4.41	+1.01	-0.86	-1.15	N/A	N/A	+149	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page